

Title: Design and Evaluation of a Prompting Tool for Token-Efficient, High-Accuracy Responses in Educational LLM Chatbots

Large Language Models (LLMs) are increasingly being deployed in applications such as chatbots to generate natural language responses to user queries. LLMs are statistical sequence models trained with self-supervised learning on massive text corpora and implemented using a transformer-based neural network architecture, which excels at handling sequences and capturing patterns in text [1].

LLMs process text by breaking it down into smaller units called tokens, which can take the form of words, subwords or characters. These tokens are mapped to multi-dimensional vectors, known as embeddings, which provide numerical representations that place tokens with similar meanings closer together, allowing the model to capture semantic relationships between the words and phrases [2]. Popular LLMs like ChatGPT have billions of parameters enabling them to predict the next token in a sequence with high accuracy [3].

The inputs for LLMs, known as prompts, are crucial for the model to produce reliable and robust outputs. Prior work has shown that LLM outputs are highly sensitive to the subtleties of prompts, and even minor alterations can lead to substantial changes or declines in model performance [4]. As LLMs grow in model size it becomes increasingly important to understand prompt sensitivity and how it affects the model's performance and behaviour [5]. Prompts can be engineered to program an LLM, customizing both its outputs and its interaction patterns within a conversation. Prompt engineering enables users to provide specific guidelines that the model treats as contextual instructions, thereby constraining style, format, and most importantly content to align outputs with the user's goals [6]. In many applications, such structured prompts can improve the relevance and accuracy of the model's responses to the user's task [4,6].

Certain prompting techniques, notably Automatic Prompt Engineer (APE), have been shown to automatically induce task-specific instructions that improve LLM performance in zero-shot settings (where the model receives only an instruction without task-specific training examples) and can discover more effective zero-shot chain-of-thought (CoT) prompts, leading to higher accuracy on challenging reasoning benchmarks including MultiArith (78.7 to 82.0) and GSM8K (40.7 to 43.0) [7].

In CoT prompting, the model is asked to generate intermediate reasoning steps, and this method has been shown to significantly improve performance on multi-step reasoning tasks compared to standard prompting. Approaches such as Active Prompting further enhance CoT performance by actively selecting informative examples, yielding higher accuracy across different reasoning benchmarks [8].

While the reliability of LLM outputs and the problem of hallucinations have been widely discussed in educational applications, most existing work focuses on high-level analysis or generic mitigation strategies rather than on dedicated prompting tools for educational chatbots. Although prompting techniques have shown significant improvements, many of these techniques require careful manual prompt design which can be inefficient and difficult for non-expert users.

This paper proposes a prompting tool that combines effective methods such as APE and Active CoT within a conversational educational chatbot, with the goal of improving the reliability and robustness of answers, including reducing hallucinations, by automatically refining suboptimal user prompts into more effective formulations that emphasise clear task constraints and step-by-step reasoning, thereby discouraging unsupported or fabricated outputs.

References:

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